



QINTERN 2026 · FINAL PRESENTATION

QuantumSentinel

A Rigorous Benchmark of Quantum Neural Networks for Zero-Day
and Adversarial Intrusion Detection in IoT Networks

Team A

Data Curation & Conformal Guarantee

Amon Koike · Mohammed Owais · Iwo Wojtakajtis

Team B

Quantum Training & Robustness

Arjun E Naik · La Wun Nannda

Team C

Intelligent Feature Selection

Kevin Wong · Maral Mahmoudi Kamelabad

Mentors: *Dr. Mohit Sajwan · Dr. Simranjit Singh*

What is QuantumSentinel?

A reproducible benchmark that asks a single, honest question: **can quantum neural networks detect the network attacks they were never trained on?**

We combine three ingredients into one pipeline:

- 1 Quantum encoding** — network flows become quantum states through a variational circuit
- 2 Prototype fidelity** — each flow is scored by how unlike every known attack it looks
- 3 Conformal calibration** — a distribution-free guarantee on the false-alarm rate

AT A GLANCE

- 3** IoT intrusion datasets (CIC-IoT2023, BoT-IoT, UNSW-NB15)
- 3** quantum architectures + parameter-matched classical baselines
- 4** evaluation axes: in-distribution, zero-day, noise, adversarial

Five questions we set out to answer

RQ1

Detection

Does the quantum model match strong classical baselines on known attacks?

RQ2

Zero-day

Does conformal calibration hit its target false-alarm rate while keeping recall?

RQ3

Disentanglement

Can it tell a disguised known attack from a genuinely novel one?

RQ4

Robustness

How does it hold up under NISQ noise and FGSM / PGD adversarial attacks?

RQ5

Honesty

Where does quantum NOT help — and what does it cost?

Zero-day attacks are the ones you never trained for

1

Single-dataset claims

Most quantum-IDS papers evaluate on one dataset and one noiseless simulator.

2

No false-alarm guarantee

Thresholds are hand-tuned; nobody bounds the false-alarm rate in finite samples.

3

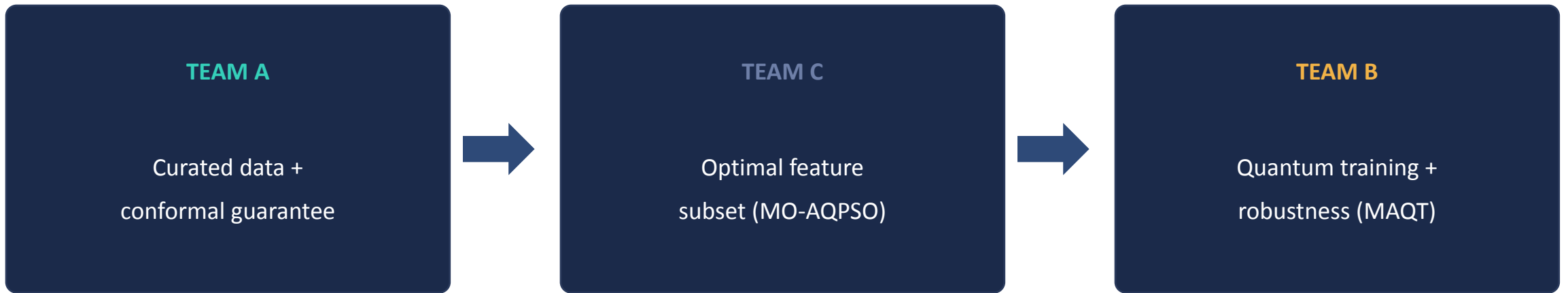
No adversarial account

What happens when an attacker disguises a known attack as novel? Untested.

OUR PROJECT

We don't try to prove quantum superiority. We rigorously test where quantum models succeed, fail, and generalize — across three datasets, realistic noise, and adversarial attacks, with statistical significance and full reproducibility.

Three teams, one pipeline



The whole pipeline is score-agnostic and frozen: real quantum scores drop in with one flag, no code change.



TEAM A

Data Curation & the Conformal Guarantee

Amon Koike · Mohammed Owais · Iwo Wojtakajtis

The data foundation & zero-day partitions

YOUR CONTENT HERE

Replace this panel with your figure, chart, table, or diagram.

WHAT TO PUT HERE

- The three-dataset benchmark (CIC, BoT, UNSW) and why TON/Edge were set aside
- Four-way split: train / calibration / test / whole held-out zero-day family
- Leakage controls: fit-on-train, known-only calibration
- One strong visual: partition table or the pipeline diagram

The guarantee holds — and we test it correctly

YOUR CONTENT HERE

Replace this panel with your figure, chart, table, or diagram.

WHAT TO PUT HERE

- The exact Beta-Binomial coverage band (not the naive $\text{FZR} \leq \alpha$ check)
- Headline: 3/3 datasets inside the band on all 5 seeds
- The coverage plot (Figure 2) tracking the diagonal
- The exchangeability-audit silent-failure finding ($p \approx 1e-209$)



TEAM B

Quantum Training, Robustness & Honest Limits

Arjun E Naik · La Wun Nannda

Training Process

ONE FULL TRAINING PIPELINE

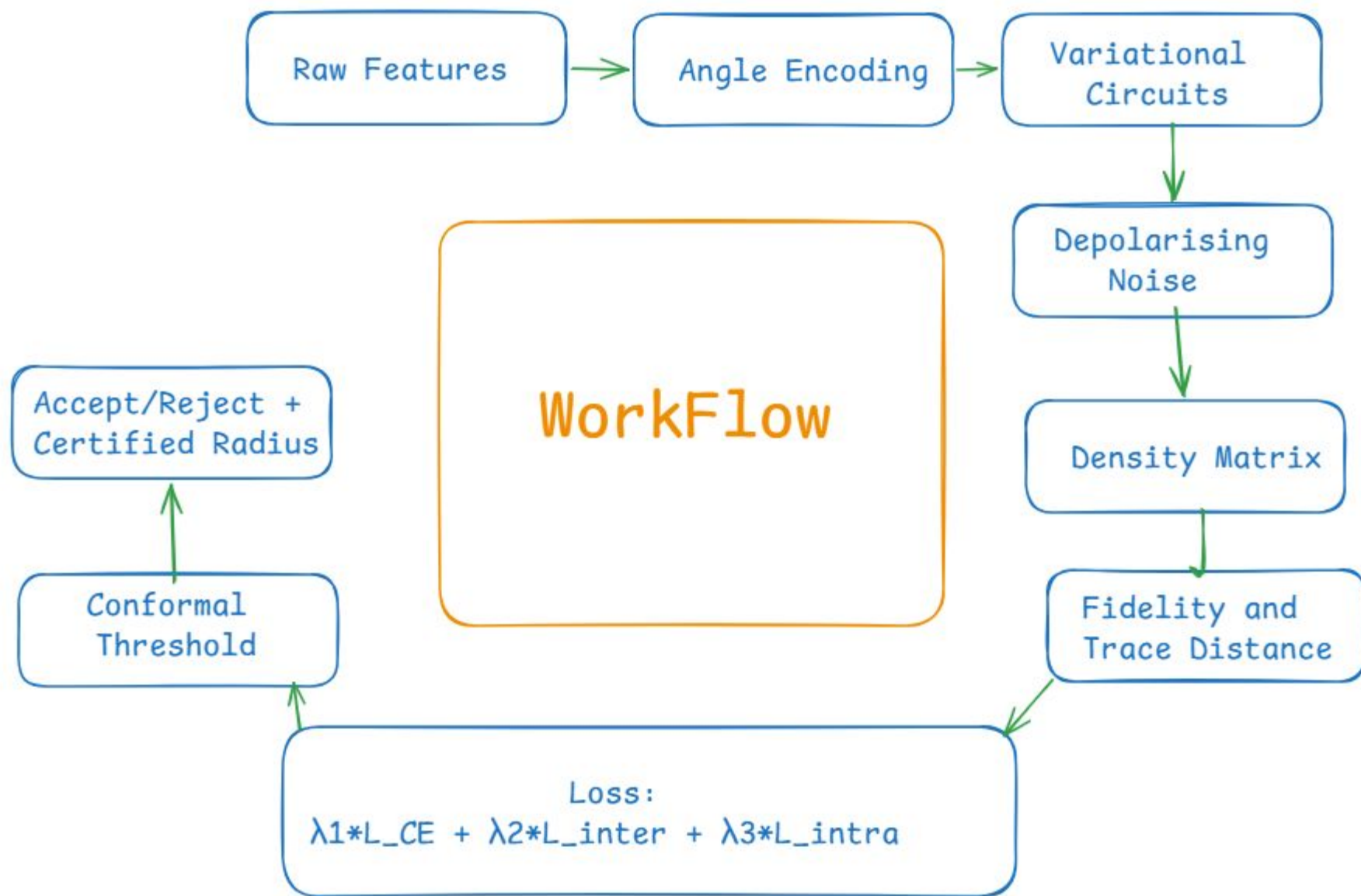
- Team C FROZEN feature handoff + Team A FROZEN data splits (train / cal / test / zeroday)
- Data Scaling
- PCA → set the number of qubits
- Further subset (for experimenting around)
- Map features → angles in $[0, \pi]$
- Build variational circuit
 - angle encode
 - data reupload
 - depolarizing noise
- Run gradient-variance / barren-plateau probe
- Train model
- Save model checkpoint

FULL TRAINING FOR ONE DATASET

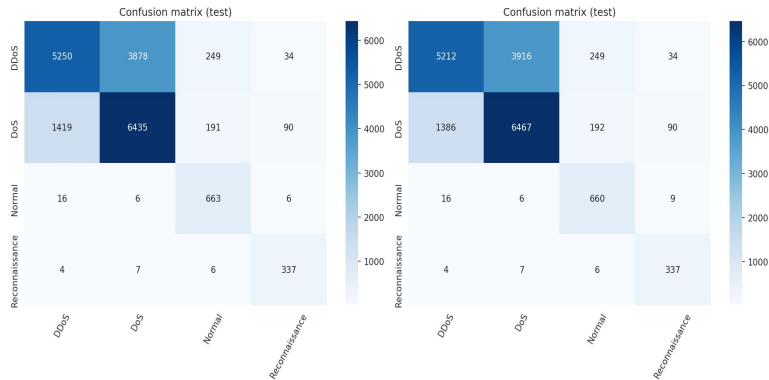
- Pick one dataset
- Do “ONE FULL TRAINING PIPELINE” for MAQT architecture
 - loss function = $L_{CE} + (\text{Lambda1} * L_{intra}) + (\text{Lambda2} * L_{inter})$
- Do “ONE FULL TRAINING PIPELINE” for VQC (plain CE) architecture
 - loss function = L_{CE}
- For each, try increasing accuracy in different settings:
 - try various lambdas
 - try different hyperparameters such as epochs, batch size, learning rate, etc.

FULL TRAINING FOR ALL DATASETS

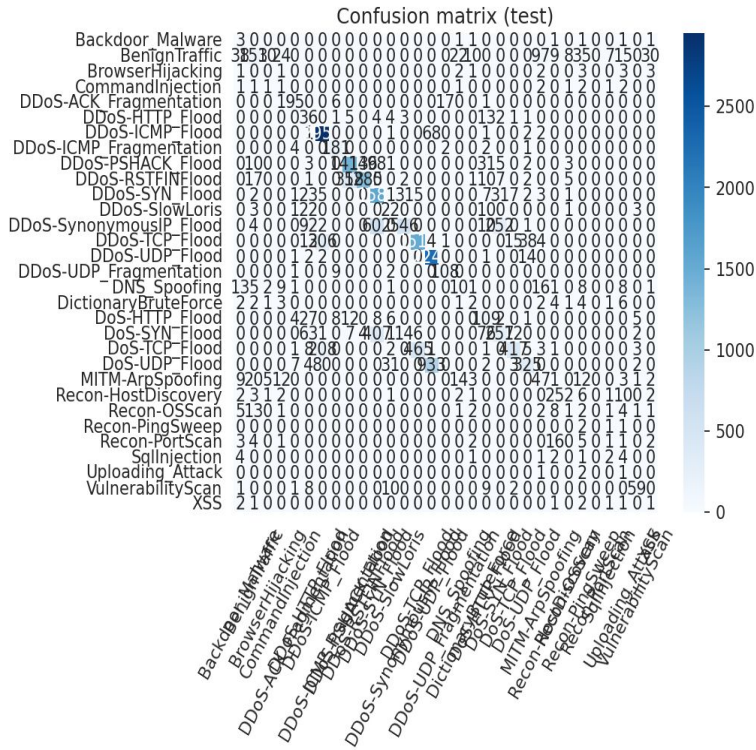
- Repeat “FULL TRAINING FOR ONE DATASET” for each of the three datasets:
 - BoT-IoT
 - CICIoT2023
 - UNSW-NB15



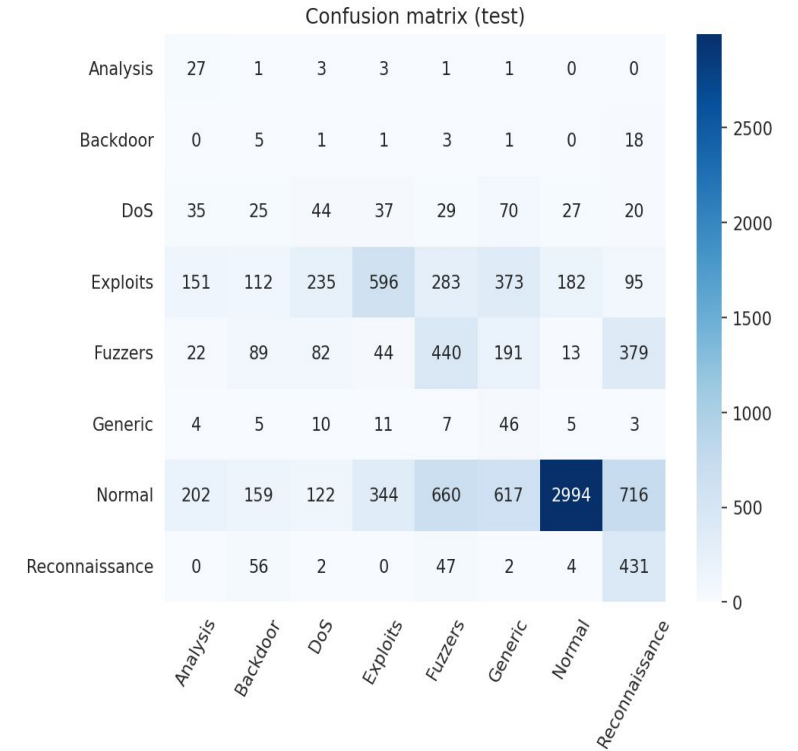
- MAQT accuracy: 68% on full test split
- VQC accuracy: 68% on full test split



- MAQT accuracy: 70% on full test split
- VQC accuracy: 70% on full test split



- MAQT accuracy: 45% on full test split
- VQC accuracy: 45% on full test split

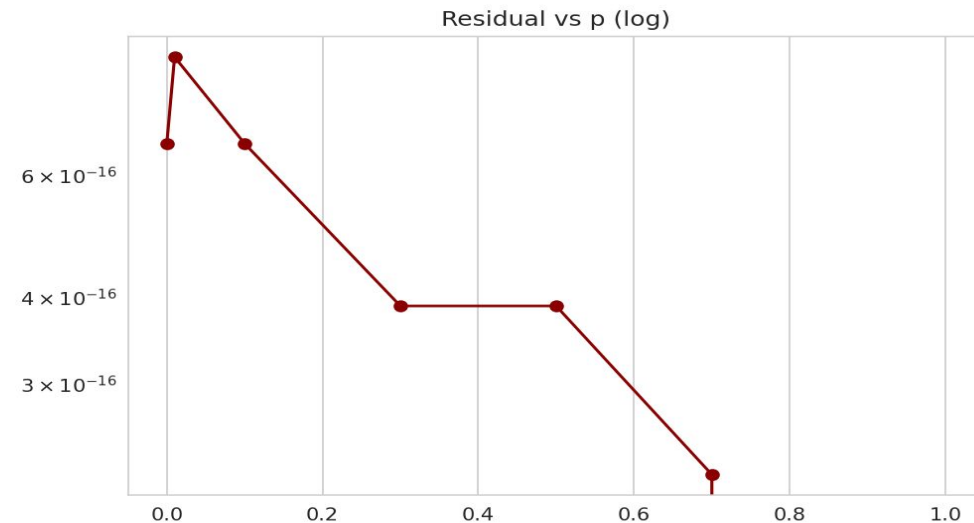
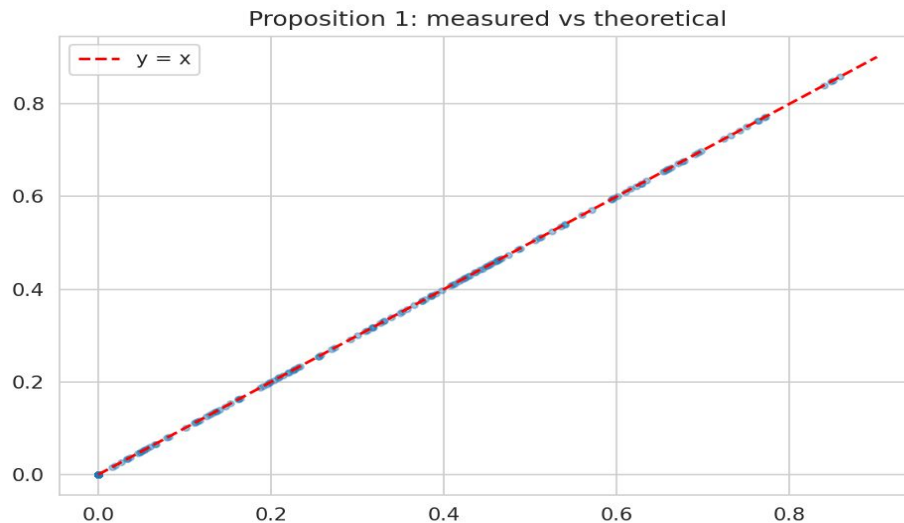


Noise Identity to Machine Precision

Proposition 1 (Noise contraction). *For the depolarizing channel Λ_p and any two states ρ, σ ,*

$$D_{\text{tr}}(\Lambda_p(\rho), \Lambda_p(\sigma)) = (1 - p) D_{\text{tr}}(\rho, \sigma). \quad (2)$$

In words: depolarizing noise shrinks the distance between any two states by exactly the factor $(1 - p)$, uniformly. It does not distort the geometry—it rescales it.



Certified Separation (Theory)

Proposition 2 (Adversarial–novelty separation). *Under assumptions (A1)–(A5), for any adversarial perturbation δ with $\|\delta\|_2 \leq \varepsilon$ there exists a rejection threshold τ such that every adversarially perturbed known sample is still accepted and every zero-day sample is still rejected, provided*

$$\varepsilon < \frac{(1 - F_{\text{out}}) - \sqrt{1 - F_{\text{in}}^2}}{2(1 - p)L_\varphi} =: \varepsilon^*. \quad (4)$$

Any threshold τ in the interval

$$\left(\sqrt{1 - F_{\text{in}}^2} + (1 - p)L_\varphi \varepsilon, \quad (1 - F_{\text{out}}) - (1 - p)L_\varphi \varepsilon \right)$$

works, and assumption (A5) together with $\varepsilon < \varepsilon^*$ is exactly the condition that makes this interval non-empty.

(A1) The encoder is Lipschitz.

There exists a constant L_φ such that for all inputs x, x' ,

$$D_{\text{tr}}(|\psi(x)\rangle\langle\psi(x)|, |\psi(x')\rangle\langle\psi(x')|) \leq L_\varphi \|x - x'\|_2.$$

(A5) The separability precondition.

The distance gap is positive:

$$\Delta := (1 - F_{\text{out}}) - \sqrt{1 - F_{\text{in}}^2} > 0.$$

(A2) Known samples are concentrated.

Every known-class sample satisfies $F_{\text{max}}(x) \geq F_{\text{in}}$: each known sample is at least this close in fidelity to its nearest prototype.

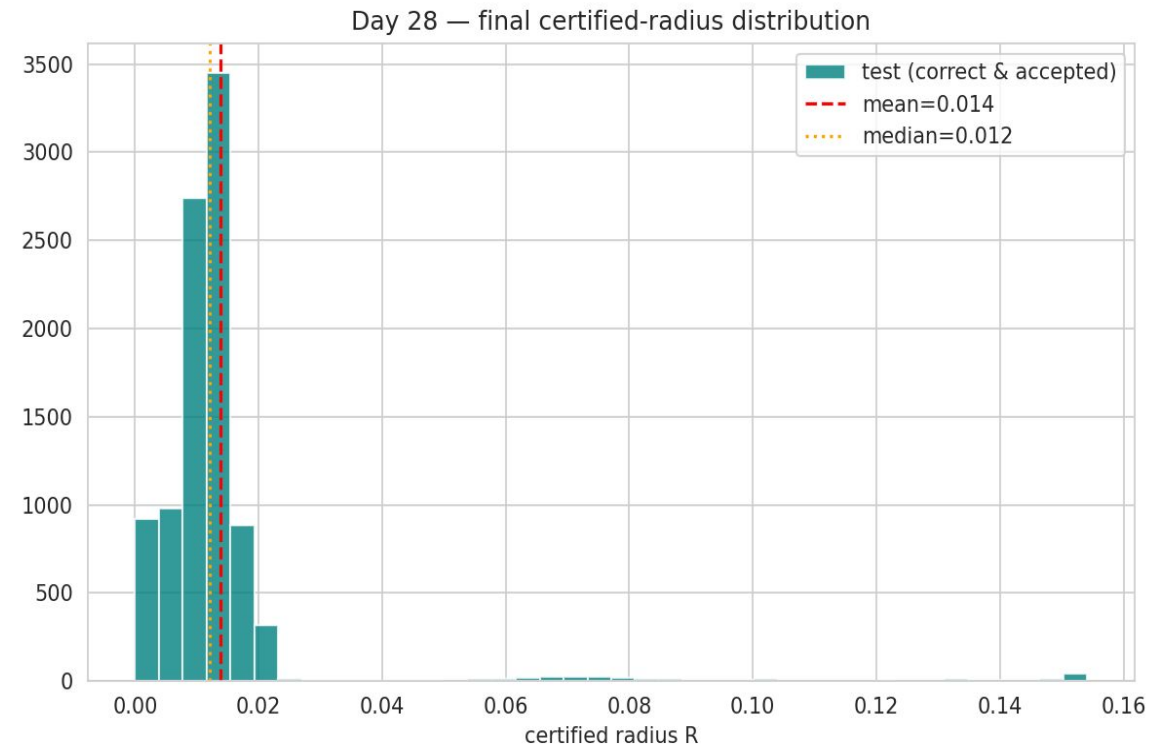
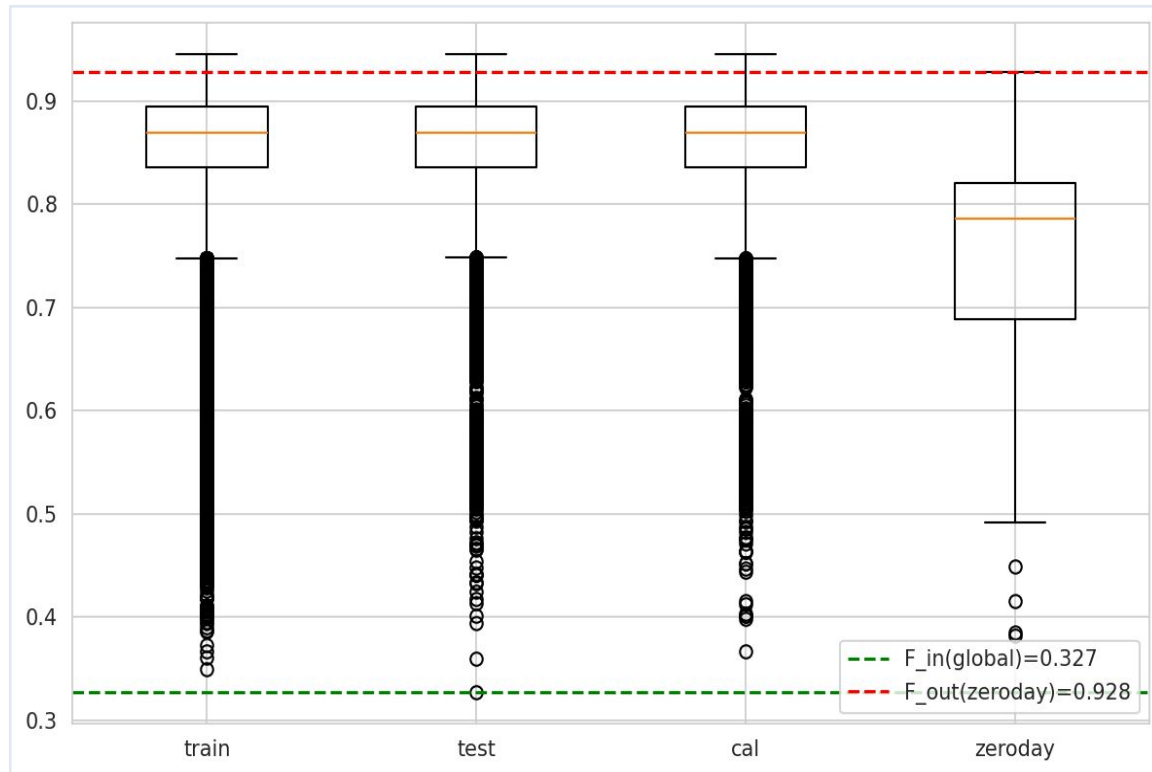
(A3) Zero-day samples are separated.

Every zero-day sample z satisfies $F_{\text{max}}(z) \leq F_{\text{out}}$, with $F_{\text{out}} < F_{\text{in}}$.

(A4) Where the fidelities are measured.

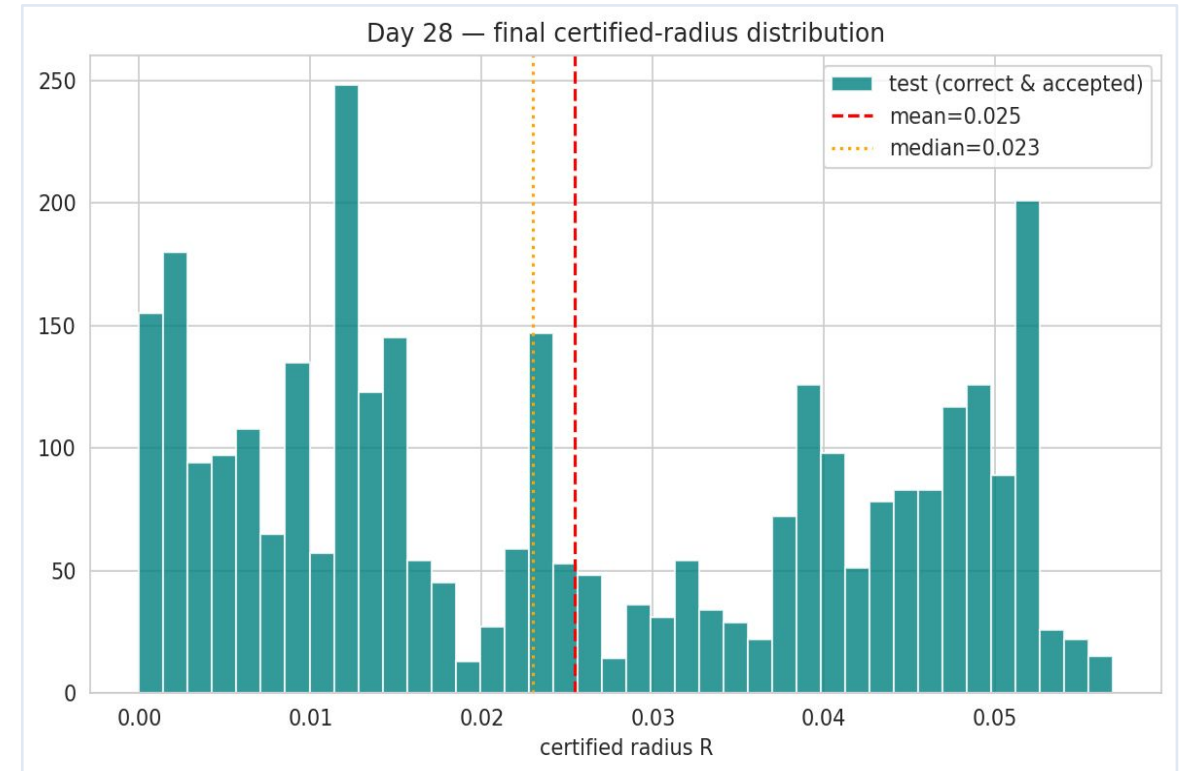
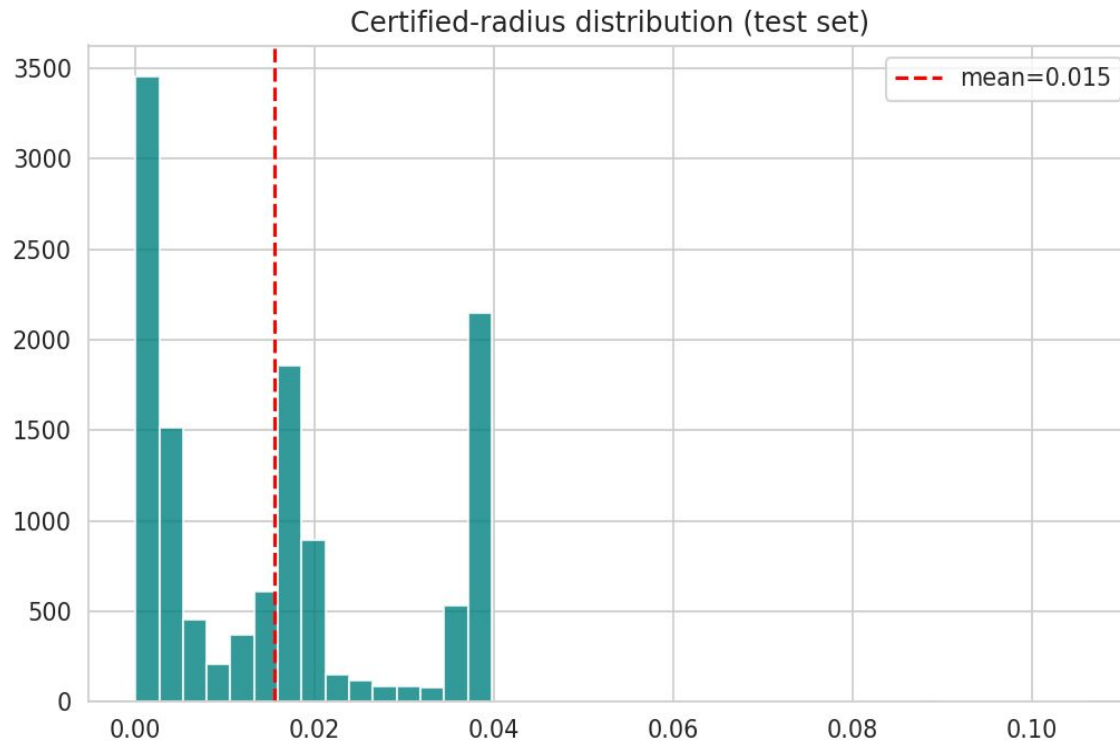
F_{in} and F_{out} are measured on the *post-channel* states $\rho(x)$, at the same noise level p used at inference—the same states the detector actually sees. This matters; see Section 3.4.

Empirical Robustness (BoT-IoT MAQT Results)

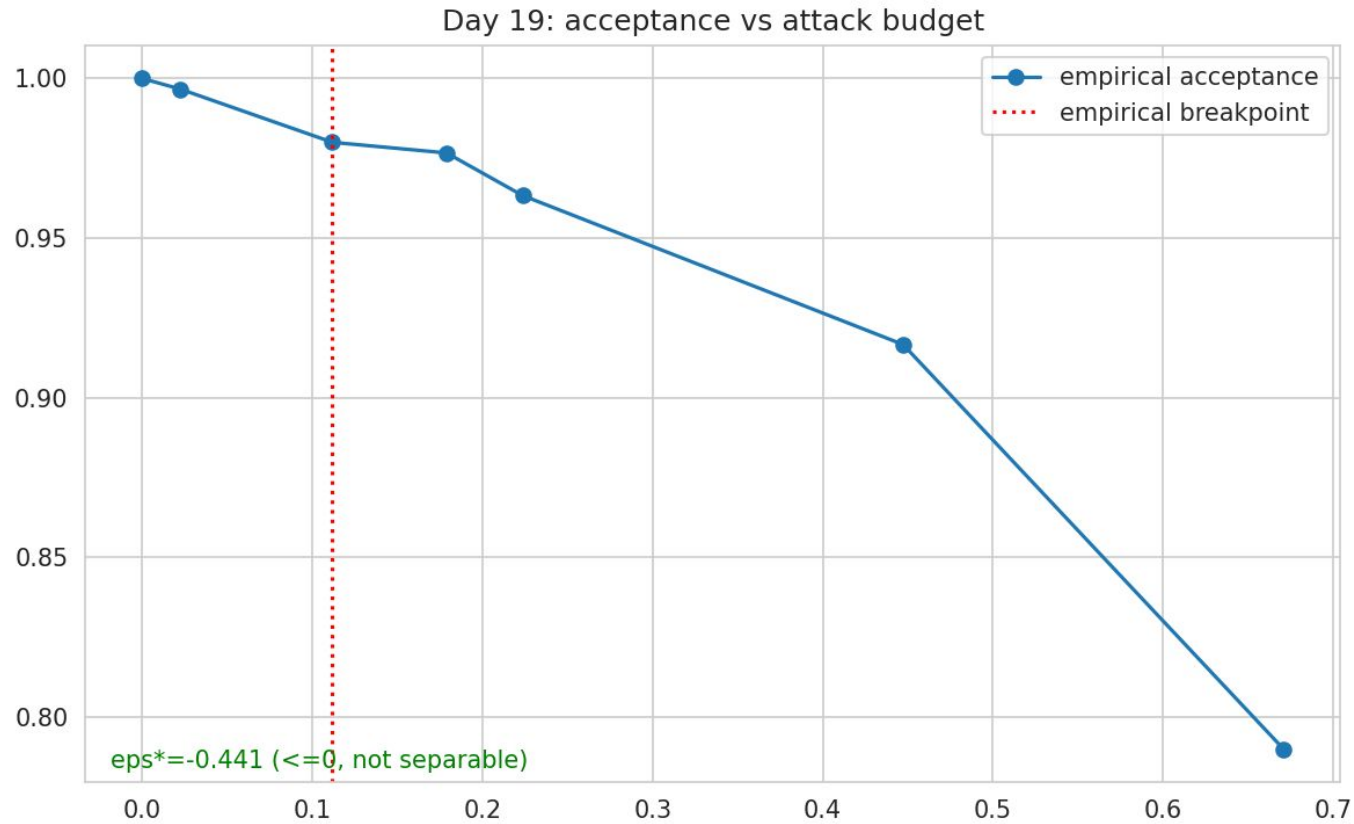


- certified separation is vacuous on real traffic (small fidelity gap)
- known samples and zero-day samples look too similar to the class prototypes

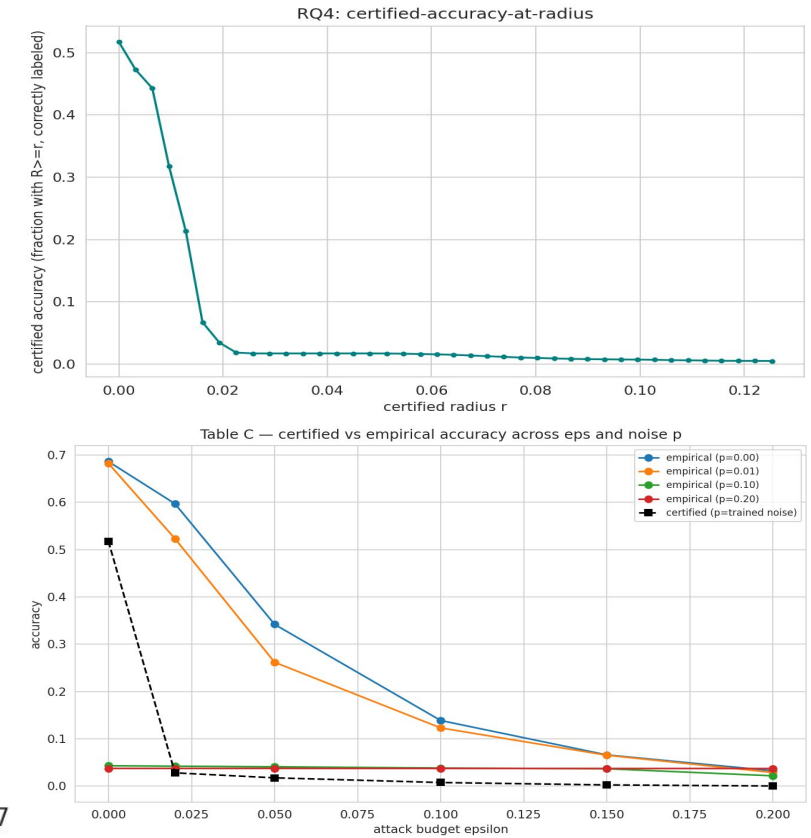
Similar Results Across Exps (CICIoT2023 & UNSW-NB15)



Empirical $\gg$ certified (sufficient $\neq$ necessary)



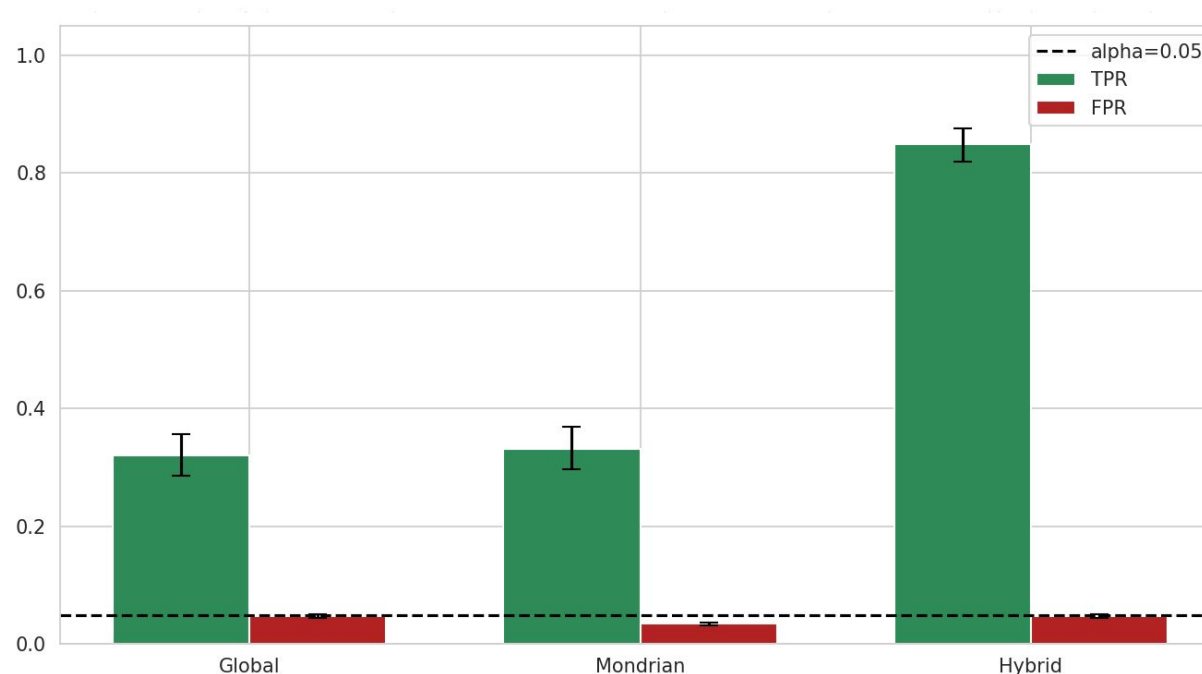
- BoT-IoT MAQT results



Zero-Day Detection: Global q vs Mondrian q vs Hybrid q

Proposition 3 (Conformal coverage). *Let $\{s_1, \dots, s_n\}$ be the nonconformity scores of the calibration set, where $s(x) = 1 - F_{\max}(x)$, and let $q = s_{(k)}$ with $k = \lceil (1 - \alpha)(n + 1) \rceil$ be the k -th smallest calibration score. If the calibration scores and the score of a fresh known-class sample X are exchangeable, then*

$$\mathbb{P}(s(X) > q) \leq \alpha. \quad (5)$$



- We set q in three ways:
 - a single global threshold (shared across all classes)
 - Mondrian thresholds (one per class)
 - a hybrid of both
- Results still vary across datasets (the shown fig. is from BoT-IoT MAQT)

Proposition 3 alpha-sweep (AUROC=0.496)

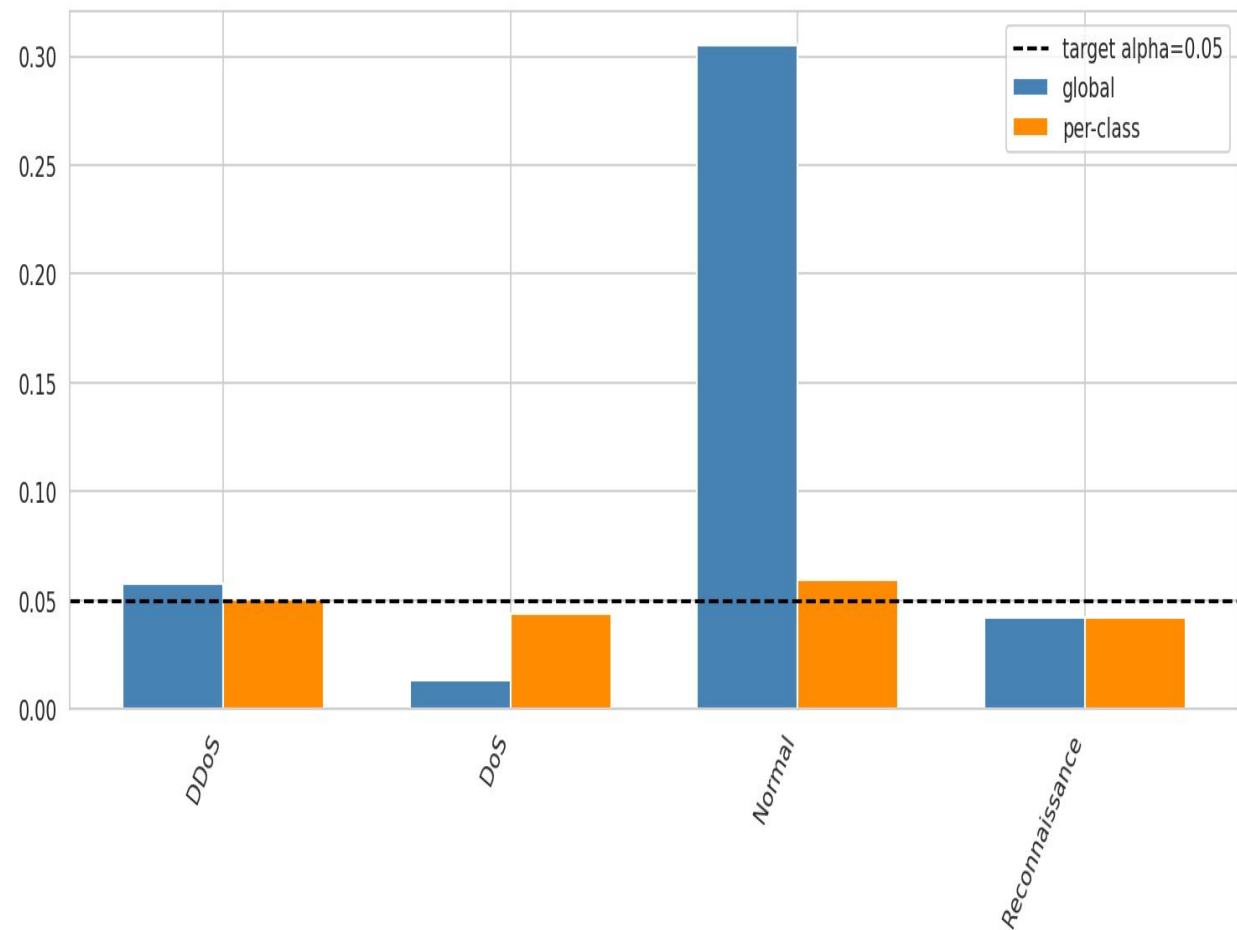
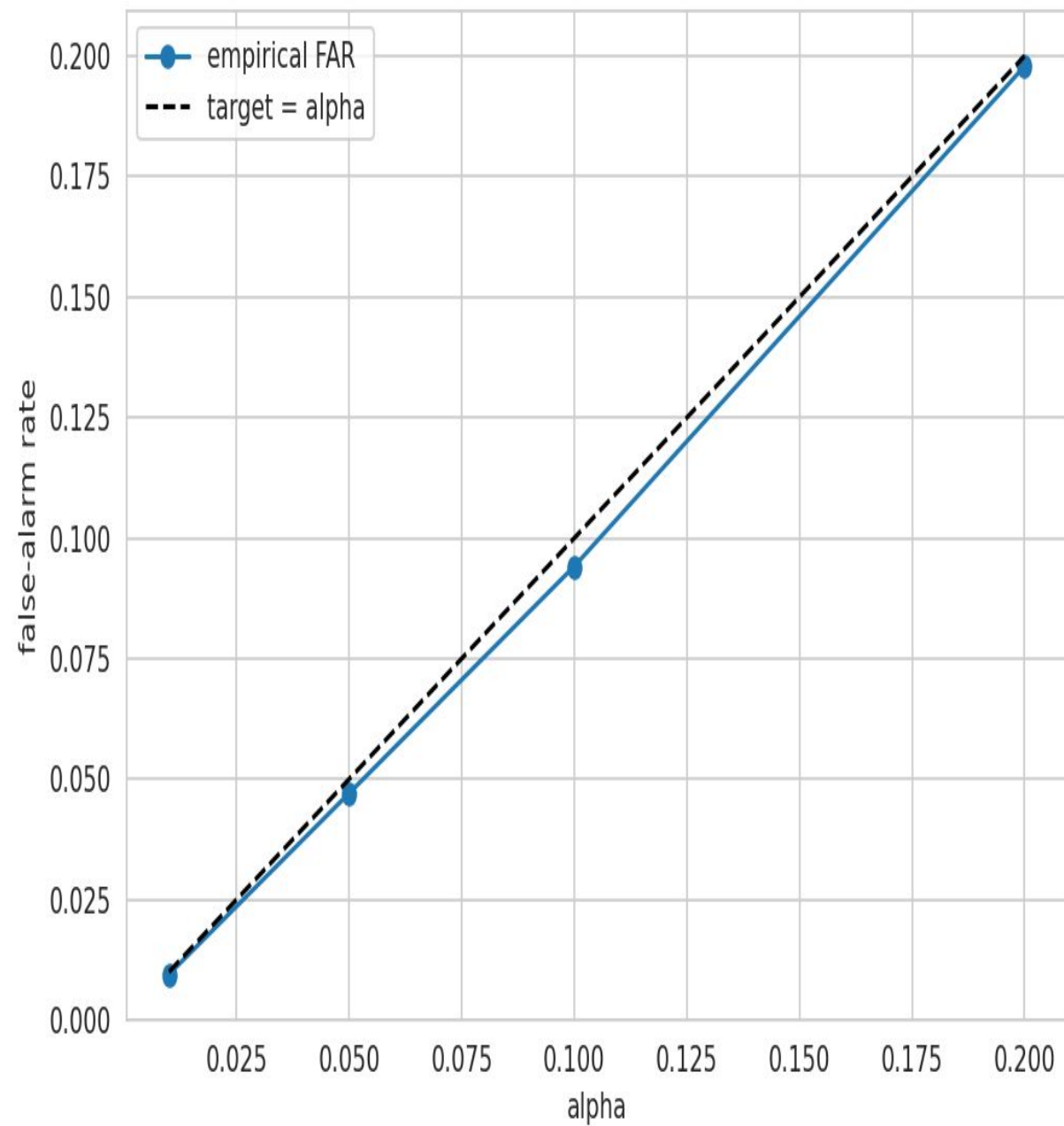


Figure 3 — Disentanglement: separation AUROC vs attack budget

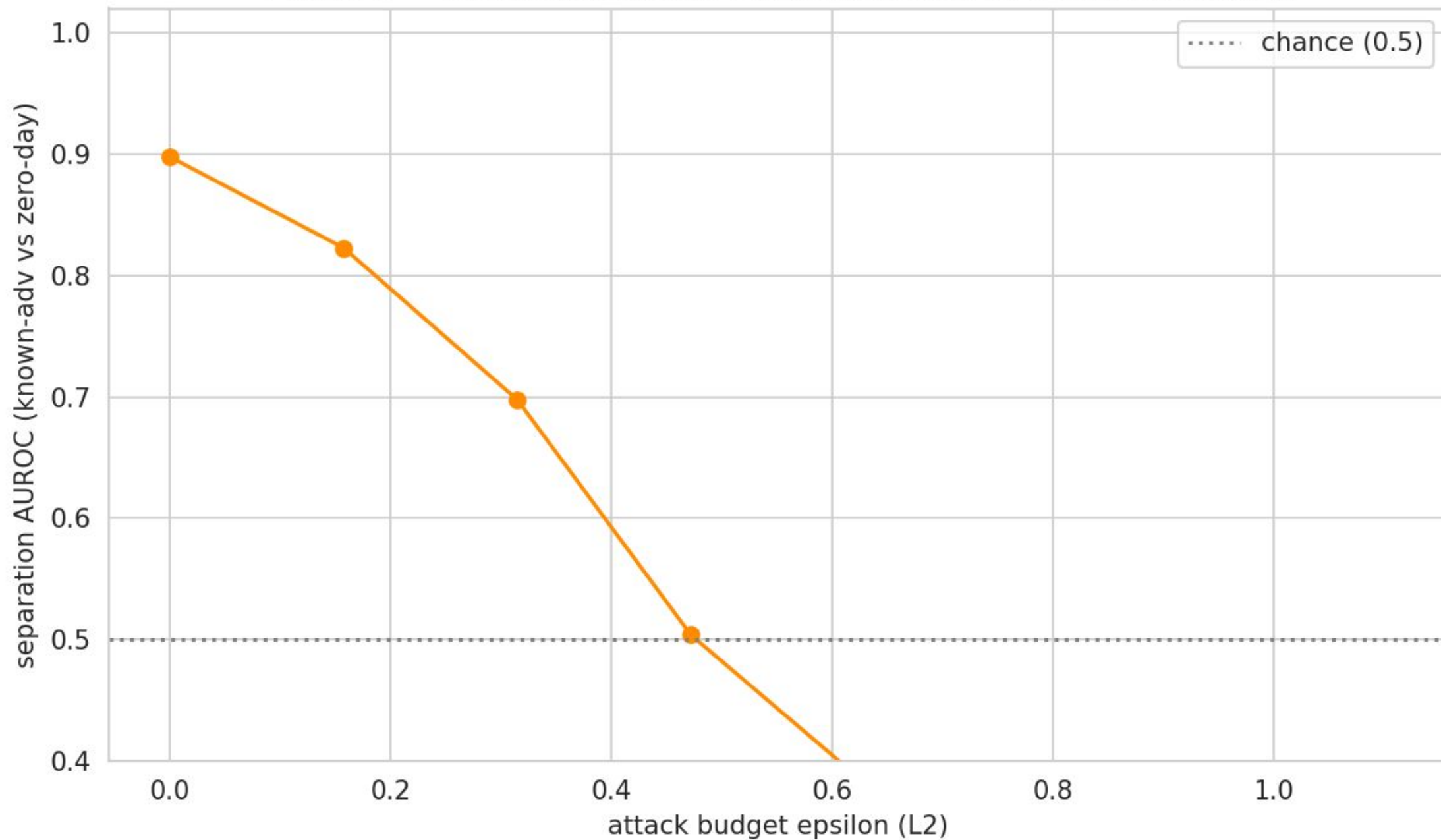
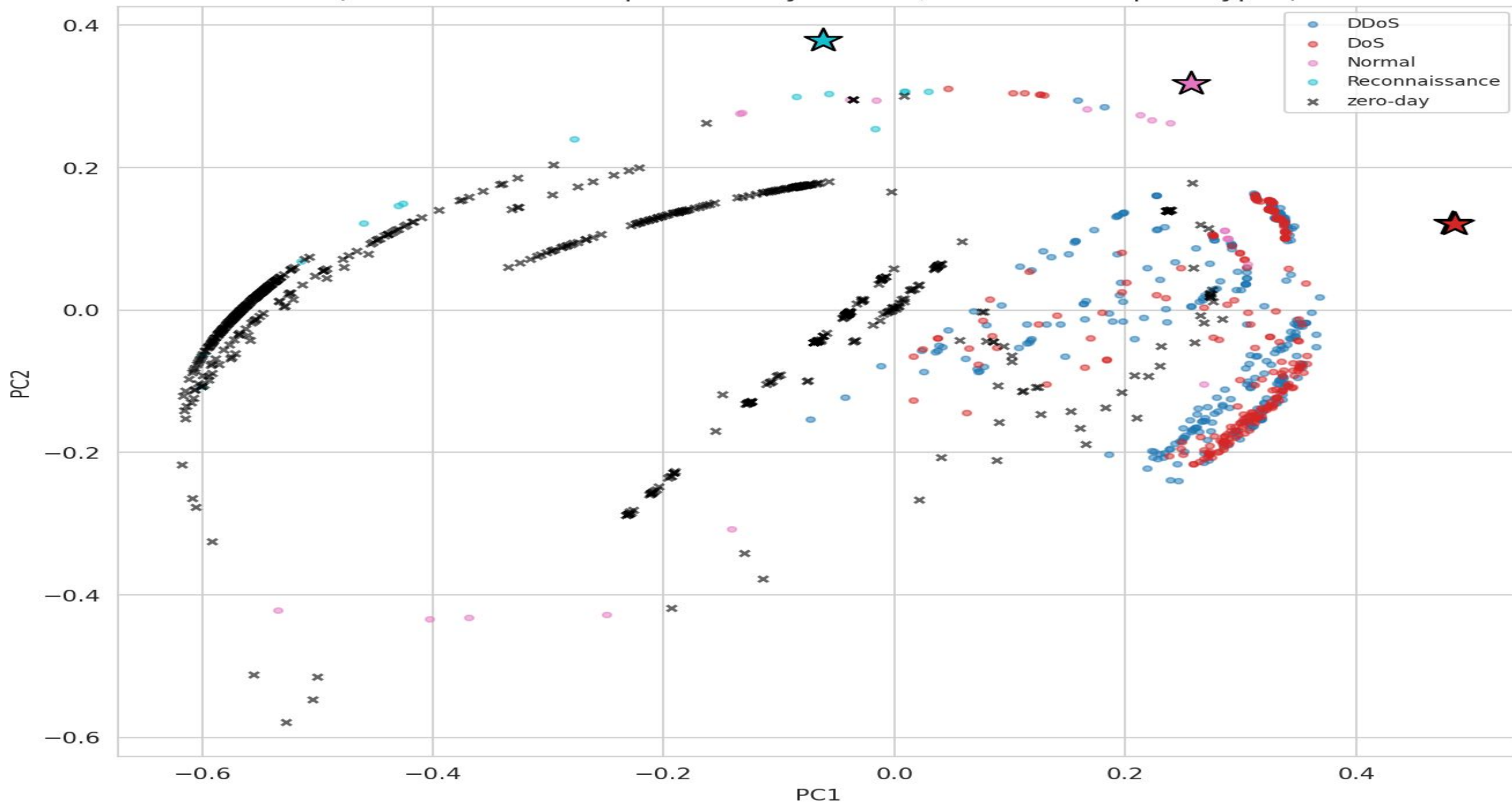


Figure 1 — Trust-region visualisation
(2D PCA of Hilbert-space fidelity vectors; stars = class prototypes)



Robustness, and an honest result

YOUR CONTENT HERE

Replace this panel with your figure, chart, table, or diagram.

WHAT TO PUT HERE

- What holds: noise identity to machine precision; conformal coverage
- The honest finding: certified separation is vacuous on real traffic — and WHY (small fidelity gap)
- Empirical robustness far exceeds the certified bound (sufficient $\neq$ necessary)
- The fix identified: a geometry-maximising loss term

MAQT: shaping the Hilbert-space geometry

YOUR CONTENT HERE

Replace this panel with your figure, chart, table, or diagram.

WHAT TO PUT HERE

- The margin-aware loss: $L_{CE} + \lambda_1 \cdot \text{intra} + \lambda_2 \cdot \text{inter}$
- Class prototypes p_c and the fidelity-gap (H1) diagnostic
- The trust-region visual: prototypes + samples in 2D
- Training setup: qubits, layers, cosine LR, seeds



TEAM C

Intelligent Feature Selection

Kevin Wong · Maral Mahmoudi Kamelabad

MO-AQPSO: multi-objective feature selection

YOUR CONTENT HERE

Replace this panel with your figure, chart, table, or diagram.

WHAT TO PUT HERE

- Four objectives: accuracy, feature count, redundancy, cost
- Benchmarked vs MI, RF, BPSO, QPSO — same classifier, fair protocol
- The Pareto-front visual (accuracy vs feature count)
- 5-seed stability (Jaccard) so the winner isn't a lucky seed

Which features, and does it help detection?

YOUR CONTENT HERE

Replace this panel with your figure, chart, table, or diagram.

WHAT TO PUT HERE

- The frozen Top-k subset delivered to Team B
- Selector → zero-day AUROC: does better selection improve novelty detection?
- Cost + stability comparison table
- The link: these are exactly the features QS-Net trains on

Results that stand up to scrutiny

3 / 3

datasets inside the exact
coverage band, all seeds

5-seed

significance: paired tests,
CIs, effect sizes

324/324

reproducibility checks
passed at sign-off

Honest

negative results reported,
not buried

- Coverage does not discriminate detectors — recall does, and it is dataset-dependent.
- The certified guarantee's limit is stated honestly, with the exact fix identified.
- Every number is traceable to a frozen, SHA-256-pinned artifact.

Why this wins: not the flashiest claim — the most trustworthy one. Rigor, honesty, and reproducibility are the contribution.



QuantumSentinel

Rigorous. Honest. Reproducible.

Thank you — questions welcome.



Team A · Team B · Team C | *Mentors: Dr. Mohit Sajwan · Dr. Simranjit Singh*